

The effect of two energy-restricted diets, a low-fructose diet versus a moderate natural fructose diet, on weight loss and metabolic syndrome param...

One of the proposed causes of obesity and metabolic syndrome is the excessive intake of products containing added sugars, in particular, fructose. Although the ability of excessive intake of fructose to induce metabolic syndrome is mounting, to date, no study has addressed whether a diet specifically lowering fructose but not total carbohydrates can reduce features of metabolic syndrome. A total of 131 patients were randomized to compare the short-term effects of 2 energy-restricted diets—a low-fructose diet vs a moderate natural fructose diet—on weight loss and metabolic syndrome parameters. Patients were randomized to receive 1500, 1800, or 2000 cal diets according to sex, age, and height. Because natural fructose might be differently absorbed compared with fructose from added sugars, we randomized obese subjects to either a low-fructose diet (<20 g/d) or a moderate-fructose diet with natural fruit supplements (50–70 g/d) and compared the effects of both diets on the primary outcome of weight loss in a 6-week follow-up period. Blood pressure, lipid profile, serum glucose, insulin resistance, uric acid, soluble intercellular adhesion molecule-1, and quality of life scores were included as secondary outcomes. One hundred two (78%) of the 131 participants were women, mean age was 38.8 ± 8.8 years, and the mean body mass index was 32.4 ± 4.5 kg/m². Each intervention diet was associated with significant weight loss compared with baseline. Weight loss was higher in the moderate natural fructose group (4.19 ± 0.30 kg) than the low-fructose group (2.83 ± 0.29 kg) ($P = .0016$). Compared with baseline, each intervention diet was associated with significant improvement in secondary outcomes. Reduction of energy and added fructose intake may represent an important therapeutic target to reduce the frequency of obesity and diabetes. For weight loss achievement, an energy-restricted moderate natural fructose diet was superior to a low-fructose diet. Copyright © 2011 Elsevier Inc. All rights reserved.